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 TOPICAL REVIEW

Blockchain-Based Reputation and Trust Management for Smart Grids, Healthcare, and Transportation: A Review

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ABSTRACT In a smart city environment, various intelligent devices, applications, and digital networks collaborate to provide technological solutions for the public good. The exponential data generated from intelligent interactions among stakeholders within the smart city ecosystem raises concerns regarding security and privacy. Maintaining data openness while safeguarding it against social engineering attacks, network breaches, and data masking threats is paramount to achieving resilience. Blockchain technology has demonstrated great promise in addressing these challenges due to its decentralized, consistent, and tamper-proof nature. Our paper explores how blockchain can address the challenges of integrity, security, and privacy in smart city operations. Despite significant research efforts, the subject remains in need of a comprehensive survey. Consequently, we conduct a study on state-of-the-art blockchain-based reputation and trust management methods within the three fundamental components of a smart city: energy, healthcare, and transportation. The trust and reputation mechanisms are analyzed to identify their strengths and limitations. The investigation reveals that the existing trust schemes are resource-constrained and encounter scalability limitations, high energy consumption, and incompatibility with existing systems.

INDEX TERMS Blockchain, consensus algorithm, cryptography, distributed ledger technology (DLT), smart city, smart grid, trust management, vehicular ad-hoc network (VANET).

I. INTRODUCTION

The concept of a smart city has surfaced as one of recent technological advancements. A smart city offers intelligent healthcare, smart mobility, smart buildings, etc. Multiple definitions of smart cities can be found in the literature [1], [2], [3], [4], [5]. A resident participates in the smart city system through his or her smart devices (e.g., smartphones or any wearable device). The intelligent devices operating in the network interact with each other in various areas of the city, such as roads, buildings, and hospitals. By optimizing operational efficiency, smart city residents receive a standardized lifestyle.

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However, there are numerous risks associated with the smart city paradigm. Despite the increased connectivity and automation, malevolent attacks are likely. One of the pressing challenges with large-scale smart city solutions is to select trusted participants. Because all of these smart devices are not trustworthy, some may maliciously harm the network or degrade the quality of service. Smart cities support critical applications demanding to protect the data and functionalities from malicious and unauthorized users [5]. Therefore, there is a critical need to create and maintain trust among network entities in the context of the network's dynamism [6]. The service providers of a smart city must adopt a trust management mechanism to ensure the legitimacy of intelligent devices.

Recently, managing trust in a network has received significant attention since it adds a security layer to ensure

that the data being exchanged in the network is valid and originates from a trustworthy source [7]. Several approaches have been proposed, including entity-based, data-based, and hybrid, to provide a suitable security and protection degree for smart city solutions [8].

Khan et al. define an energy-efficient trust management scheme for smart cities using an Intrusion Detection System (IDS) [9]. The trust among smart devices is modeled using opinion triangles in subjective logic. If a node's reputation falls below a certain threshold, it is considered malicious and is quarantined. However, this approach is resource-intensive. The research by Haseeb et al. overcomes this limitation [10]. It introduces an optimal data relay algorithm to attain less congested and traffic-aware routing paths for trusted nodes.

Trust management solutions based on edge and cloud computing have recently been proposed for smart cities to minimize computing and transmission powers. However, these are exposed to network anomalies and vulnerable to various attacks. Xu et al. utilized the SPEA2 algorithm for privacy preservation of edge nodes [11]. Al Hamadi et al. adopted another approach to support smart services provided as a cloud utility [12]. End users automatically measure one-to-one service ratings and credibility ratings. The beta reputation system by Josang is defined to calculate the service rating and taste similarity [13]. A user subjectively selects the best or most trustworthy service provider based on its performance metrics. Wang et al. also leveraged edge computing to define a trust management mechanism for smart cities [14]. The mechanism is based on the concept of a dynamic black-and-white list. Devices with high trust values are added to the whitelist (trustworthy). In contrast, devices with low trust values are added to the blacklist (untrustworthy). Kurdi et al. employ peers' reputations and Service Level Agreements (SLAs) for trustworthy peer-to-peer communication among federated clouds [15]. The peer nodes evaluate the trustworthiness of other peers based on subjective logical opinions.

Trust management is a promising approach to establishing security for the heavily resource-constrained devices within a smart city infrastructure [9]. Nevertheless, the existing trust schemes pose challenges due to their limited functionality, efficacy, and scalability. Many challenges result from the large-scale deployment of smart devices in a smart city that need to be considered:

- Most trust management techniques defined for smart cities are centralized and rely on a single server or a centralized authority to store and manage identity attributes and authorization policies [16].
- Infiltration from malicious devices is a significant challenge facing smart cities, as it can potentially disrupt or compromise their services and functions [17].

- Identification data and authorization policies are stored within a single database owned by an external trusted third party [18].
- Massive overheads incur when several nodes simultaneously communicate with the central server.

Decentralized trust management systems are gaining attention as they address centralization issues in traditional trust management methods. Blockchain technology (BCT) is being leveraged to tackle central management problems. For instance, Asif et al.'s research [19] introduces a secure authorized access system for smart city resources using blockchain, the Authorization for Constrained Environments (ACE), and the Object Security Architecture (OSCAR). Wang et al.'s work [20] proposes a reputation-based Proof of Work (PoW) consensus for trust management in smart cities.

Blockchain is considered a breakthrough technology [21] and is built on the idea of Distributed Ledger Technology (DLT) as its foundation [8]. Hypothetically, blockchain is similar to a traditional database that requires atomicity, consistency, isolation, and durability (ACID) properties to be satisfied. The significant difference between the two is the "consensus" algorithm that decides whether a new block is legitimate to insert or not to insert into the blockchain. A blockchain-based system combines a P2P network with a shared ledger, cryptography, and a computing platform to manage the network's transactions, record-keeping, and security.

This paper aims to systematically review blockchain-based trust management methods used in smart cities. We examine the current state-of-the-art techniques in three critical applications of smart cities: smart grids, smart healthcare, and smart transportation. Our focus is on decentralized trust management models based on blockchain technology. By doing so, we provide a comprehensive overview of the existing research in this area and identify future research directions. Our paper makes the following contributions:

- 1) An introduction to blockchain technology that covers its implementation as a distributed ledger for privacy and security applications.
- 2) A comparison of customary consensus algorithms, focusing on five key aspects: fault tolerance, performance, degrees of decentralization, scalability, and recourse consumption.
- 3) A comprehensive review of value-based decentralized trust management models with Blockchain for Smart Grids, Healthcare, and Transportation.
- 4) Discussion of major challenges and issues when implementing blockchain technology trust management systems.

The remainder of the paper is structured as follows. Section II describes the adopted research methodology. Section III provides an overview of blockchain technology with its applications, implementation, and challenges. It also discusses the potential of BCT in the intelligence-sharing domain. Section IV presents an overview of a trust management system and various blockchain consensus mechanisms. The

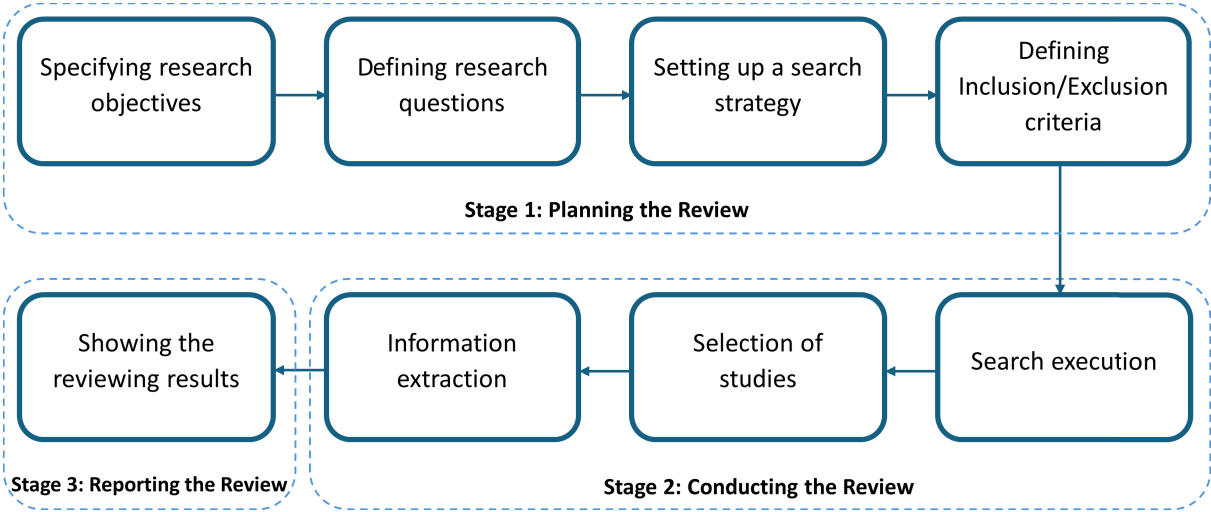


FIGURE 1. Systematic review methodology.

following sections review blockchain trust management techniques in smart city applications. Section V presents the trust management studies for smart energy systems. Likewise, the research conducted in the healthcare sector is introduced in Section VI. Blockchain-based trust management systems for smart transportation are couched in Section VII. Research challenges are highlighted in Section VIII. Finally, Section IX concludes this paper.

II. METHODOLOGY

This section presents the methodology that we adopted to carry out the review. The review methodology is organized into three stages, as depicted in Figure 1. Each stage addresses different aspects of blockchain-based trust management, from study selection to data analysis and synthesis of findings.

A. STAGE 1: PLANNING THE REVIEW

This is the first stage of our systematic literature review. It comprises (1) Specifying the research objective, (2) Defining research questions, (3) Setting up a search strategy, and (4) Defining Inclusion/Exclusion criteria.

1) RESEARCH OBJECTIVE

This survey aims to systematically review the state-of-the-art blockchain-enabled reputation and trust management mechanisms within a smart city reference.

2) RESEARCH QUESTIONS

This review paper aims to answer the following research questions (RQ):

- **RQ1:** What are the fundamentals of blockchain technology, including its operating principles, and how does it affect data sharing in various domains?
- **RQ2:** Could blockchain be a viable platform for sharing intelligence and information between government agencies and the private sector?

TABLE 1. Outcome of databases search.

Database	Search 1	Search 2	Search 3
Google Scholar	5280	8340	1310
IEEE	85	104	39
Scopus	54	122	37

- **RQ3:** What are blockchain technology’s most commonly employed trust management models and consensus algorithms?
- **RQ4:** What theoretical frameworks or models are commonly employed to study blockchain-based trust management in smart city contexts?
- **RQ5:** What are the open research challenges and potential future research directions for investigating blockchain-based trust management implementations in smart city environments?

3) SEARCH STRATEGY

Research Databases: We selected the relevant research papers after careful consideration from three popular databases: IEEE Xplore, Google Scholar, and Scopus.

Search Stings: The search included keywords such as “trust management” and “blockchain” in the titles, abstracts, and keywords of research papers. To identify more relevant articles, we also searched for “certificate,” “authentication,” and “safe communication.” We further refined our search by using specific keywords related to smart city applications, including “Smart Grids,” “Healthcare AND IoMT,” and “VANET AND Vehicles.” In the second phase of our search, we combined these keywords with “trust” and “blockchain” to broaden our search and identify more relevant papers.

- *Search 1:* Blockchain trust management smart grids
- *Search 2:* Blockchain trust management healthcare IoMT
- *Search 3:* Blockchain trust management VANET vehicles

4) INCLUSION AND EXCLUSION PROTOCOL

We defined inclusion and exclusion criteria for selecting relevant articles.

Inclusion Criteria:

- Year Range: 2018 to 2023
- Document Type: Journal article and conference paper
- Language: English

Exclusion Criteria:

- Journal articles and conference papers before 2018
- Papers with no citations
- Non-English manuscripts
- Overlapping papers

B. STAGE 2: CONDUCTING THE REVIEW

We have conducted a systemic review of 51 research articles that fulfill our inclusion criteria.

1) SEARCH EXECUTION

The search phase was very straightforward. We queried the above-mentioned research databases for the specified terms in the title, abstract, and keywords of technical and survey papers.

2) SELECTION OF STUDIES

We followed a two-stage screening strategy. In the initial phase, we manually reviewed the extracted papers' titles, abstracts, and keywords. We ensured that the selected papers were relevant to the subject, i.e., blockchain-enabled trust management. We also made sure that there was no repetition of articles. Sometimes, the title and abstract do not provide a well-defined reflection of the paper. Hence, we reviewed the full text of the articles and chose the relevant/non-relevant paper based on our defined inclusion and exclusion criteria in the second phase.

Due to the novelty of Blockchain-assisted smart grids, we encountered many challenges in finding relevant papers. Our search revealed that "trust management" was often used interchangeably with "authentication." Despite targeting high-citation papers, we had to settle for lesser-cited research in areas like smart grids and smart healthcare due to the limited availability of literature in these fields.

C. INFORMATION EXTRACTION

We categorized the papers in each domain. We then comprehensively studied the selected research papers in full detail to logically segregate these articles based on adopted methodology, key features, strengths and limitations, etc.

D. REVIEW RESULTS

In this subsection, we present the statistics related to our search outcomes.

After a comprehensive search, we selected 51 research papers (including bibliographic reviews) that focused on blockchain-based trust management approaches for smart grids, healthcare, and transportation.

TABLE 2. Distribution of the collected references.

Topic	Percentage of references
Trust management in the energy sector	33%
Trust management in healthcare	32%
Trust management in VANETs	35%

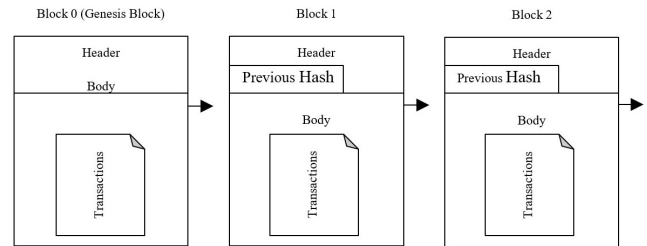


FIGURE 2. Logical layout of blockchain.

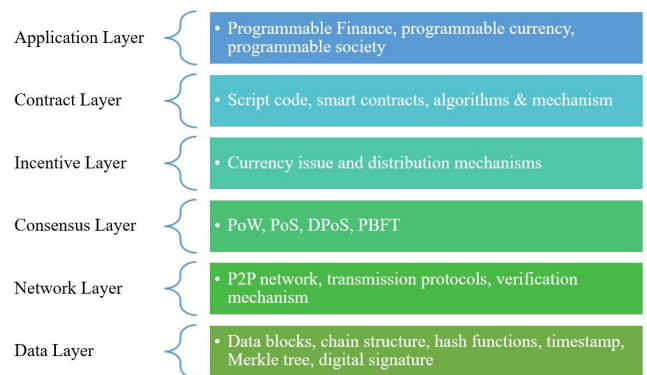


FIGURE 3. Blockchain layered architecture [25], [26].

Table 1 presents the outcome of our selection from the three databases in each studied sector. Of these papers, 76% were published in refereed journals, while 24% were published in the proceedings of tier 1 international conferences. Table 2 shows the distribution of the selected references.

E. REPORTING THE REVIEW

III. BLOCKCHAIN ECOSYSTEM AND ITS APPLICATIONS

This section addresses **RQ1** by concentrating on BCT, its architecture, implementation, functioning, and technological challenges, and **RQ2** by describing how blockchain is a viable platform for sharing intelligence and information between government agencies and the private sector.

A. BLOCKCHAIN TECHNOLOGY

Blockchain is fundamentally a peer-to-peer network that provides secure execution of transactions without the need for a trusted third party [22]. As previously discussed, blockchain is a backward-linked list of blocks chained together [23], as illustrated in Figure 2. Each block includes a header, certificates, and its previous block's hash [24]. The block header includes the block's ID, the generator's ID, the signature of the block, the hash of the block, and the timestamp. The block certificate includes the certificate's ID, node's ID, node's public key, a hash of the certificate, and

timestamp. Every new block contains the previous block's hash, which contains the previous block's hash, and so on.

Blockchain comprises six layers [25], [26] as shown in Figure 3. It works by creating a tamper-resistant chain replicated all over the blockchain network. A node employs a digital signature using private-key cryptography for creating a transaction [27]. A transaction is secured by an encryption code composed of transaction information, sender, and receiver.

Blockchain is categorized into three types based on permissions given to network nodes:

- 1) *Permissioned (private)*: These blockchains are managed by users eligible to authenticate transactions. The outside user may be allowed to read transactions, but only members of the closed group have "Write" privileges [28]. Hyperledger Fabric is a business framework and one of the most known permissioned blockchains [29]. GemOS [30] and MultiChain [31] are also private blockchain platforms.
- 2) *Permissionless (public)*: This type of blockchain network allows every transaction to be visible to all nodes, and any node within the network can participate in the consensus process to validate a transaction. In the case of a consortium blockchain, all the participating nodes can take part in the consensus process, but only the selected nodes are designated to confirm a block [27].
- 3) *Consortium (hybrid)*: A consortium blockchain can be thought of as a "semi-private" setting [32]. It is a hybrid version of both public and private blockchains. It is granted to a group of approved individuals/organizations commonly associated with an enterprise. Ethereum supports the building of consortium blockchains.

Table 3 compares different blockchain infrastructures.

While BCT is still developing, it has yet to fully address its challenges, making it a relatively new and evolving area of innovation. Despite numerous advantages, blockchain suffers from bottlenecks in design, implementation, protocols, etc. Primarily, blockchain faces the challenges of trust management and data sharing [33].

- Design Complexity [34], [35]
- Governance and Regulations [27]
- Privacy and Security [36]
- Interoperability [37], [38], [39]
- Fork Problem [40]
- Scalability [41], [42], [43]
- Storage Capacity [44], [45]
- Computational Complexity [46]
- Energy Consumption [47], [48]

B. APPLICATIONS OF BLOCKCHAIN

In recent years, BCT has shown its adaptability in various fields, and it has the potential to revolutionize nearly all domains. Nevertheless, it is probable that shortly,

TABLE 3. Comparison of blockchain infrastructures.

Attribute	Public	Consortium	Private
Access	Anyone	Multiple selected organizations	Single organization
Participants	Permissionless, anonymous identities	Permissioned, known identities	Permissioned, known identities
Decentralization	Decentralized	Partially decentralized	Centralized
Security & Consensus	PoW or PoS	Pre-approved participants, voting/multiparty consensus, PBFT, PoET, PoA	Pre-approved participants, voting/multiparty consensus, Ripple
Immutability	Immutable	Partially immutable	Alterable
Transparency	Transparent	Partially transparent	Opaque
Transaction execution speed	Slow	Lighter and faster	Lighter and faster
Scalability & Flexibility	Poor	Good	Superior
Popular platforms	Bitcoin, Ethereum, Litecoin, Factom, Dash, Blockstream	Hyperledger, Ripple, R3	Multichain, Blockstack, Bankchain, GemOS

organizations will create their unique blockchain platforms and protocols to prioritize effectiveness and efficiency.

Blockchain applications range from banking, education, supply chain management (SCM), and healthcare to public and social services. Typical applications of blockchain technology include digital currency, reputation systems, cybersecurity, and many more [49].

1) SMART CITY OPERATIONS

The development of smart cities indeed benefits from blockchain technology in terms of intelligent transport systems, smart construction, energy, education, and security. Blockchain has the potential to give unique solutions to the Internet of Vehicles (IoV). The blockchain attributes can be exploited to efficiently develop a self-organizing data management platform to support various decentralized applications. Public blockchain networks can also implement auction mechanisms for resource allocation in the IoV. Blockchain can be seen as a promising technology for smart mobility and electric vehicles (EVs) as it supports secure, reliable, and decentralized energy trading.

2) BITCOIN AND FINANCE

Blockchain is considered a breakthrough with use cases ranging from globally deployed cryptocurrency systems like Bitcoin to smart contracts, smart grids, and so on [25]. Blockchain was first proposed as Bitcoin's distributed ledger to alleviate the double-spending problem. At present, its

applications have gone far beyond Bitcoin or crypto. The primary stimulus to digital ledger technology has come from its applications as a medium for exchanging crypto assets, but its transforming potential is much higher. The researchers believe that blockchain has the potential to revolutionize many fields, leading to three generations of blockchain, i.e., Blockchain 1.0, focused on digital currencies; Blockchain 2.0, targeting digital finance; and Blockchain 3.0, aimed at creating a digital society [50].

3) PRIVACY PRESERVATION

Several studies propose enabling blockchain technology for smart contract distinction, anomaly detection, and malicious contract identification. The data stored on the blockchain cannot be modified, making it a legitimate disruptor for industries like healthcare, banking, and cybersecurity [50]. It is also believed that blockchain will benefit other industries because it can store tamper-proof data and manage a vast trail of records efficiently [27]. One of the applications that seems most promising in connection with blockchain is the application of smart contracts. In securities trading and post-trading, a distributed ledger creates a shared digital database that records the ownership of securities and custody of assets, accessible to authorized parties only.

C. BLOCKCHAIN FOR SECURITY APPLICATIONS

BCT was created to ensure several inherent security attributes, such as consistency, tamper-resistance, pseudonymity, resistance to several attacks, double-spending problems, and forking. From a security perspective, the Blockchain is created and maintained using a peer-to-peer overlay network and is secured through intelligent and decentralized utilization of cryptography with crowd computing [25]—the fundamental security properties of blockchain stem from both cryptography advances and BCT implementation itself.

Recent research suggests that the most security-oriented blockchain applications include the Internet of Things (IoT), data storage and sharing, network security, personal user data, navigation, and internet services [51]. Recently, Blockchain has attracted much attention in cyberspace and is considered a good fit for providing security and privacy.

Numerous security applications can enhance their capabilities by integrating blockchain technology. Secure storage and recovery systems are typical innovative applications of Blockchain technology. Data sharing between any application and the end users can be made safe and legitimate by string and passing through a blockchain [51]. Similarly, end-users of IoT-enabled wearable devices can benefit from the technology by having their preferences encrypted and stored on the Blockchain, making them retrievable only by the same user. A blockchain Domain Name Server (DNS) supersedes a standard DNS; it is decentralized and brings in new functionalities, like countering censorship by concerned authorities and preventing DNS spoofing.

Likewise, DL-based access control for web applications can be proposed by harnessing the intrinsic properties of Blockchain that robustly enforce access control policies without depending on a single trusted party [52].

In a content delivery network (CDN) setting, motivating a node to share its computational resources with other participants within the network can be challenging. This inhibits the growth of the network and undermines the power of defense systems. Blockchain, as a breakthrough distributed system, rewards the participants who participate in improving a security system [53].

Blockchain can also be exploited to grant secure access to Wi-Fi hotspots. To achieve this, users are authenticated with the credentials stored in a blockchain as signed transactions. Digital signatures prove that credentials are held for the targeted access point.

As Blockchain ensures the integrity and transparency of data storage and retrieval, it has the potential to be employed for the intrusion detection domain as well [33]. The intersection of Blockchain with intrusion detection enhances the security in IoT network traffic and cloud to detect malicious nodes effectively, empower trust management, and protect critical data.

D. BLOCKCHAIN FOR INTELLIGENCE SHARING

Blockchain technology enhances information security through its intrinsic cryptography, giving rise to high data integrity as each transaction is coupled with user information on who initiated the transaction and other related tasks. Interestingly, these transactions can be tracked with full transparency as the blockchain maintains a complete history of the transactions. At the same time, the distributed nature of the blockchain promotes the immutability, traceability, and verifiability of an individual's data. Moreover, in a traditional cyber threat intelligence (CTI) system, the threat intelligence information can be interfered with [54].

Blockchain technology offers promising features for maintaining a traceable and tamper-resistant ledger for intelligence sharing. Information or data gathered by the intelligence community, like the judiciary, security, legislature, and commercial code registries, is susceptible and requires efficient data storage and sharing [55]. However, centralized data systems are exposed to exploitation, which could lead to data breaches and subsequently affect a country's national security or strategic interests. The researchers argue that a decentralized, flexible, and scalable framework can overcome this issue. Blockchain can be leveraged for data management to tackle such issues. Prior studies suggest integrating blockchain to develop secure data management systems [45], [56].

However, a blockchain-inspired threat intelligence sharing model can overcome the issue through its key characteristics, i.e., consensus, decentralization, and traceability. Also, decentralized intelligent network edges (DINEs) face many challenges when discovering and accumulating edge

knowledge due to their complicated nature [51]. Edge knowledge is an output of edge computational intelligence (ECI) in heterogeneous IoT networks. Adversaries may eavesdrop on the edge knowledge or temper the data, leading to Denial of Service (DoS) and man-in-the-middle attacks. A blockchain-inspired framework may be employed to preserve edge knowledge sharing in IoT.

In most countries, regulatory compliance demands the privacy of patients' data as it contains one's medical history and laboratory test results. Hence ensuring the security and privacy of health-related data [55]. Nevertheless, several challenges are also attached to EHR systems because the centralized server models are vulnerable to single-point attacks and sometimes suffer from malicious insider attacks. In such cases, patients lack access to their medical history, hindering doctors from providing adequate treatment. This can lead to poor prognosis and deterioration of quality of life. The Health Insurance Portability and Accountability Act (HIPAA) of 1996 emphasizes the importance of patient data accuracy, confidentiality, integrity, audit, and accountability. If blockchain is incorporated into the EHR systems, it can significantly help ensure their security. This will ultimately encourage individuals and organizations to share data and facilitate audit and accountability. To summarize, a distributed ledger of shared information like blockchain is the ultimate need of threat intelligence philosophy.

IV. TRUST MANAGEMENT

A. OVERVIEW

This section focuses on the concept of trust, including its definition, types, and classification, from a blockchain-based perspective to address RQ3.

In the communication domain, trust management is a broad concept. It consists of information collection, trust-related content storage, trust calculation, trust maintenance, and decision-making [58], [59]. It provides a novel solution to security problems in heterogeneous, open, distributed, and dynamically changing network environments [60], [61].

Trust management is one of the most crucial features for providing a secure and reliable service to the residents of a smart city. It is the first line of defense among the digital systems' security dimensions, i.e., trust, traceability, privacy, integrity, and availability [62]. Identity management, access control, and mutual authentication are the main components of a trustworthy smart environment to ensure secure communication and data exchange between communicating entities. Otherwise, these environments will become vulnerable to various security risks, such as identity theft, unauthorized access, and data alteration [63].

Wang et al. segregate the trust realm into decision models, evaluation models, and management models based on their design goals [64]. Here, we focus only on trust management models. The existing trust management systems (TMS) are divided into four classes: recommendation-based,

prediction-based, reputation-based, and policy-based [61], [65].

Trust calculation (generally referred to as inference) is the process of calculating or speculating the degree of trust of a node in the evaluation data collected [66]. Typically, Bayesian, fuzzy reasoning, maximum likelihood, weighted average, and grey reasoning are the most common BC trust calculation methods described in the literature.

According to the trust management mode, a trust model can be classified into centralized, decentralized, and semi-centralized [61] as shown in Figure 5.

- 1) *Centralized models*: In the centralized TM, the trust management module is deployed on a high processing server or a central authority [64], [67]. Although a centralized model gives a complete bird's eye view of the network system, it suffers from a single point of failure. Additionally, its architecture lacks scalability, and the cost of maintaining a centralized server is very high at the same time.
- 2) *Decentralized models*: Unlike a centralized trust model, a central authority does not exist here. Trust is calculated by every node or a set of nodes themselves [68]. Although decentralized models eliminate the single point of failure, these inevitably impose massive computation and energy overheads on the system as individual nodes perform trust management computations.
- 3) *Semi-centralized models*: This class of trust management works as a combination of centralized and decentralized models. Here, a centralized authority exists, which is responsible for the overall network management, and simultaneously, some distributed nodes perform trust-related tasks. Semi-centralized models overcome most of the disadvantages of both centralized and decentralized mechanisms.

Blockchain has been extensively used within the trust management paradigm to realize dynamic authentication of users and devices in a network, where trust degrees are stored in the blocks [5]. Wei et al. describe the critical components of blockchain-enabled trust management for an IoT service-oriented architecture [57] as shown in Figure 4.

Blockchain creates a layer of trust between un-trusted parties to enable secure and trusted records and transactions to occur. For that matter, it employs different cryptographic methods [49]. The following techniques are commonly used to implement trust management systems:

- Certificate Generation
- Cryptography (ECC, ECDSA, SAS, SHA-256, SSE, PKI, etc.)
- Bayesian Inference
- Fuzzy Logic
- Trust Values (rating/ reputation)

Several techniques, such as digital signatures, encryption, and certificates, serve as security countermeasures to provide privacy and integrity [69]. Addressing certain challenges

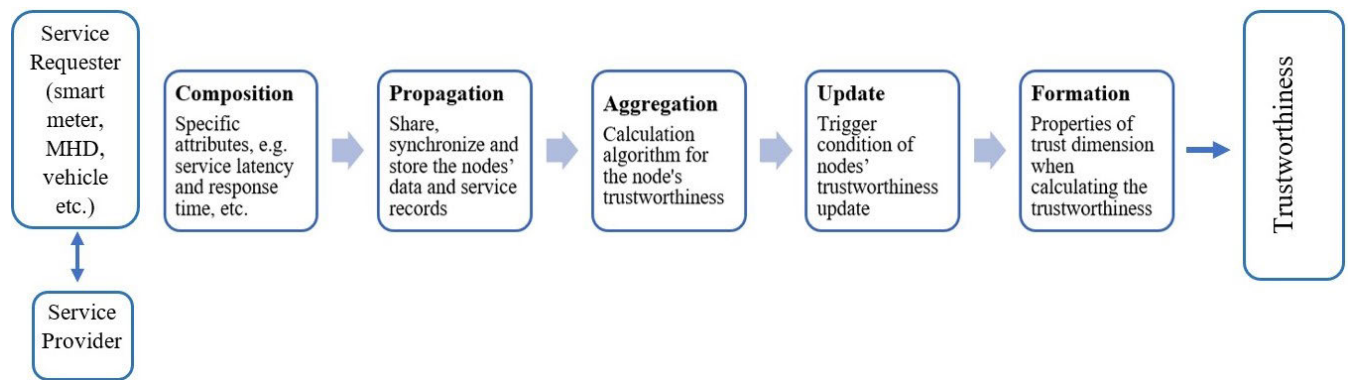


FIGURE 4. Generic model of trust management [57].

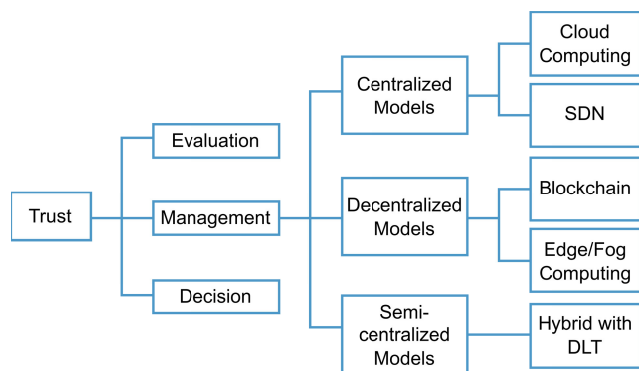


FIGURE 5. Trust taxonomy [64].

is vital for effectively deploying public key cryptography, Diffie-Hellman, and ECC in vehicular communication systems.

1) DIFFIE-HELLMAN

Diffie-Hellman is an asymmetric public-key encryption method that securely exchanges cryptographic keys between two parties over a public channel [70]. It is widely used for secure communication but raises several challenges when applied in VANETs.

As the number of vehicles increases, managing and retaining communication among vehicles becomes difficult, requiring more computational resources. Additionally, the overhead from key exchange affects performance and scalability [71]. Diffie-Hellman is an asymmetric cryptographic protocol. This is more secure than symmetric approaches because it uses two keys. Generation of two keys causes high energy, memory, and time consumption [72]. Diffie-Hellman provides high security as the key is never directly transmitted over the channel. However, this encryption technique is highly vulnerable to the man-in-the-middle attack [72]. Integration of complementary cryptographic methods with Diffie-Hellman can enhance its potential for implementation in VANETs.

2) ELLIPTIC-CURVE CRYPTOGRAPHY (ECC)

ECC is an asymmetric encryption, also known as public-key cryptography, having a public and a private key. The

ECC algorithm is more suitable for vehicle communication because it requires smaller encryption and decryption keys [73], [74].

Although ECC is more efficient than traditional cryptographic methods, the computational requirements can still be significant for resource-constrained devices, such as in-vehicle units (IVUs) and Roadside Units (RSUs) [69], [75]. Cryptographic operations in ECC consume a significant amount of energy, which is a concern for battery-powered components in vehicles. ECC, while efficient compared to other methods, still consumes more energy than simpler algorithms [70]. The energy consumption is directly proportional to the execution complexity of an algorithm. Hence, when big data is produced in vehicular networks, it affects the operational efficiency of ECC [76]. Consequently, the energy consumption is increased. The ECC-based authentication scheme ensures the message's integrity by signing the messages and verifying the signatures [77]. However, it is still vulnerable to side-channel attacks or quantum computing threats. Furthermore, establishing trust in a decentralized vehicular network can be challenging. With ECC, balancing security with energy efficiency and computational overheads is critical.

3) PUBLIC-PRIVATE KEY CRYPTOGRAPHY

Public-private key cryptography, also known as asymmetric cryptography, is a fundamental technology in modern cryptography. It involves using a pair of keys: a public key, which is shared openly, and a private key, which is kept secret. These cryptographic schemes have various problems concerning their performance and efficiency [72].

Public-private key encryption is not well-suited for dynamic environments like VANETs, as the certificate management results in computational overheads. On-board units (OBUs) in vehicles may have limited computational resources, making it challenging to perform intensive cryptographic operations. Cryptographic operations can be energy-intensive, which concerns battery-powered OBUs and roadside units (RSUs). Also, it is difficult for a vehicle with electrically operated components to manage and store abundant public/private key pairs and the related certificates [72].

This imposes huge storage requirements for each vehicle as well. Public-private key cryptography is essential for ensuring secure communication in VANETs. Still, it suffers from eavesdropping, man-in-the-middle attacks, and Sybil attacks due to frequent joining and leaving of vehicles in the network [78].

Zero-knowledge proof (ZKP) is a cryptographic technique with three intrinsic properties; zero knowledge, completeness, and soundness. A Zero-knowledge Succinct Non-Interactive Argument of Knowledge (zk-SNARK) is a Zero-Knowledge protocol to prove some computational facts about the data without the disclosure of the data [79].

ZKP arguments are the main building blocks for blockchain-based privacy solutions [80]. In a blockchain-enabled security environment, ZKP can verify whether the prover has enough transaction volume without leaking any private transaction data [81]. ZKP is a promising interactive verification protocol for the secure exchange of digital assets, secure remote biometric authentication, anonymous verifiable voting, secure auction, etc. [81].

Jiang et al. incorporate a non-interactive zero-knowledge proof scheme with blockchain for privacy preservation of vehicles [82]. The proxy re-encryption mechanism is employed to manage the symmetric key of the data. Zhang et al. propose a secure information-sharing scheme based on blockchain and zk-SNARK [83]. The system uses an interplanetary file system (IPFS) to securely store and share task information in a fully decentralized manner. Another research utilizes the extended version of zero-knowledge, i.e., Simulation Extractable Zero-Knowledge SNARKs (SE zk-SNARKs) to achieve anonymity and conditional privacy in vehicular networks [84].

Sharma et al. define an access control mechanism for EHRs using zero-knowledge proof [85]. Blockchain is utilized to store the patients' data. The patient's identity is authenticated using zero-knowledge proofs and access is granted via proxy re-encryption. The research by Luong et al. introduces a privacy-preserving mutual authentication mechanism for healthcare networks [86]. The protocol uses the zero-knowledge succinct non-interactive argument of knowledge (zk-SNARK) and blockchain for privacy preservation. zk-SNARK and the anonymity property of Ethereum allow patients to access healthcare services with complete anonymity and confidentiality.

B. TRUST-BASED CONSENSUS MECHANISMS IN BLOCKCHAIN

A consensus algorithm is a solution designed to address the "Byzantine Generals' problem." As all the nodes are considered equal in a blockchain network, the network is termed a distributed network. Such a scenario requires a facilitation mechanism to establish trust between the interacting nodes [87].

With the advent of blockchain technology, multiple consensus algorithms have been proposed depending on

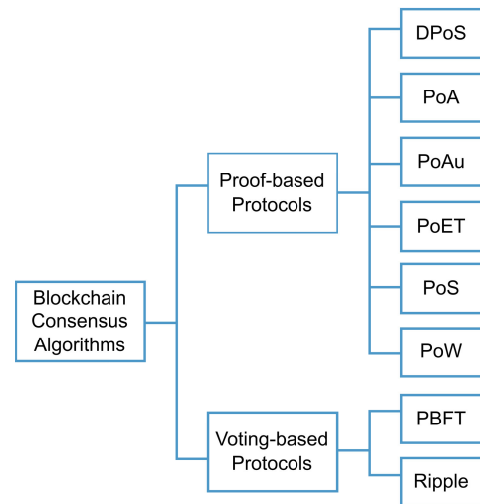


FIGURE 6. Trust-based consensus algorithms.

their usability. However, an efficient consensus algorithm is resistant to attacks, fault-tolerant, performant, and accessible to all network participants.

Consensus protocols in blockchain are broadly categorized as either proof-based or voting-based [88], [89]. The respective consensus protocols under each category are reflected in Figure 6. A proof-based consensus algorithm relies on the notion that among the numerous nodes connected to the network, the node that conducts adequate verification is granted the privilege of adding a new block to the chain and is rewarded [90]. Voting-based algorithms are based on Byzantine fault-tolerance and have solid mathematical proofs to ensure security and stability.

1) POPULAR CONSENSUS PROTOCOLS

The following are the most widely used consensus algorithms:

- *Delegated Proof of Stake (DPoS)*: DPoS consumes less energy and leads to faster transactions than PoW and PoS. It implements a one vote per share mechanism that enhances the number of processed coins [89]. As the nodes vote for randomly selected witnesses to preserve consensus, they are incentivized or penalized based on their generated blocks and accomplishments. DPoS limits decentralization as it employs many validators to reach a complete consensus.
- *Proof of Activity (PoA)*: A research work by Bentov et al. combines PoS and PoW to render PoA consensus protocol [91]. PoW gives the decision-making power to entities performing computational tasks, while PoS gives the decision-making power to entities with a stake in the system.
- *Proof of Authority*: Proof of Authority (PoAu) is a variant of PoS for permissioned blockchains, and it belongs to the family of Byzantine Fault Tolerance (BFT) algorithms [92]. This algorithm supersedes a lighter message transmission scheme as compared to BFT [93].

A PoAu network has high throughput, is highly scalable, and has almost zero processing overheads. However, it tends toward strong centralization due to a small number of entities controlling the whole network.

- *Proof of Elapsed Time (PoET)*: Intel Corporation introduced the Proof of Elapsed Time (PoET) consensus method in 2016 to improve energy efficiency and eliminate the waste of resources [94]. The participants solve a hash calculation problem like PoW. However, the participants have no competitive relationship; the winner is selected randomly. This consensus protocol highly depends on dedicated hardware to restrict cooperation and decentralization.
- *Proof of Stake (PoS)*: The Proof of Stake algorithm supersedes PoW regarding high energy consumption. PoS aims to stake peers' economic share in the network, and one of the validators (miners) is chosen to publish a block onto the blockchain. Here, a validator is selected in a pseudo-random fashion instead of working to solve a computational puzzle. Due to this, block finality in PoS-enabled blockchains is faster than in PoW-enabled blockchains.
- *Proof of Work (PoW)*: The most known and trusted consensus algorithm is Proof of Work (PoW). Dwork and Naor originally proposed PoW to address the problem of spam emails [95]. It is considered to be the first blockchain consensus protocol and is based on computational load, requiring miners to find a solution to a puzzle. Several cryptocurrencies utilize a variant of this protocol. The main feature of the PoW is that a node completes a specific task to get the reward and win [87]. PoW requires high computational power, lacks scalability, and poses long latency for transaction confirmation over the network.
- *Practical Byzantine Fault Tolerance (PBFT)*: This algorithm was first introduced in 1999 to reduce the complexity of the original Byzantine fault tolerance algorithm from exponential to polynomial [96]. It is the most widely used consensus algorithm as it ensures the normal consistency of each node and the order of up-link transactions [97]. However, PBFT is inappropriate for use in SCM, energy trading, etc. [98].
- *Ripple*: Also known as the Proof of Concept (PoC) consensus algorithm. Financial institutions typically employ Ripple to facilitate faster cross-border transactions. The validating nodes preserve a set of trusted nodes in the form of a Unique Node List (UNL). The UNL must maintain agreement among 80%

Table 4 and Table 5 present the comparative analysis of state-of-the-art proof-based and voting-based consensus algorithms based on their features and application scenarios.

2) ALTERNATIVE PROTOCOLS

Over time, multiple other consensus protocols have been devised based on the traditional algorithms depending upon their utility. Delayed Proof of Work (dPoW) was designed

by Supernet as part of the Komodo project to protect against the double-spending problem [99]. It is a hybrid consensus method, leverages the security feature of another blockchain, and prevents 51% of attacks. Proof of Energy (PoE) was explicitly designed to administer peer-to-peer energy trading using distributed ledger technologies [100]. Smart contracts are deployed with PoE to regulate energy transactions with less power consumption. Casper is an alternative to PoS. It incorporates dynamic validator sets and a correct-by-construction forking scheme [93]. The consensus mechanism adds PoS on top of PoW as a supplementary layer to ensure finality. Nevertheless, again, it is unable to cater to 51% of attacks. Another derivative of PoS is Proof of Claim (PoC) and was initially deployed by CLAIM Coin [93]. It eliminates the proportional incentive mechanism of the former PoS and enhances network security.

In addition, some authors have developed new archetypes that depart in other ways from the traditional consensus notions. Proof of Cooperation (PoC) is a new consensus algorithm proposed by Khalid et al. [101]. As the name suggests, it enhances the cooperation among network entities for validation and mining of blocks. With PoC, all agents have an equal chance of being selected as the combiner. Prabhakar and Anjali proposed a trust-dependent Consensus (TCON) consensus algorithm [102]. Trusted cluster heads (CHs) validate blocks. A mathematically formulated trust metric decides the number of cluster heads that must validate a new block and the number of transactions to be validated. TCON is proven to save computational power and communication energy. Proof of Endorse Contracts (PoEC) consensus is proposed by Cheng et al. [103]. It is a cross-blockchain consensus mechanism and resolves trust between a supervisory blockchain (permissioned, private, or consortium) and a supervised blockchain (permissionless public). Recently, Praveen et al. proposed a Proof of Majority (PoM) consensus algorithm to eliminate resource-intensive tasks leading to a reduced carbon footprint [104]. PoM allows nodes to become miners without providing any resource-intensive tasks. The algorithm outraces PoS and PoW regarding throughput, decentralization, and entry barriers.

V. BLOCKCHAIN-BASED TRUST MANAGEMENT IN SMART GRIDS

This section addresses **RQ4** by surveying the literature on trust management models implemented through blockchain technology in smart grids.

Smart grids are significant in modern energy infrastructure due to intelligent electronic management. In recent years, there have been several potential blockchain-based applications in the smart grid domain [112]. Some smart grid applications using blockchain are related to power generation, consumption, transmission, and distribution. Table 6 shows the classification of the literature studied under distributed trust management in smart grids.

TABLE 4. Comparison of proof-based consensus algorithms.

Property/Feature	DPoS	PoA	PoAu	PoET	PoS	PoW
DLT Type	Permissioned/Permissionless	Permissioned	Permissioned/Permissionless	Permissioned/Permissionless	Permissionless	Permissionless
Decentralization	Semi-centralized	High	Centralized	High	Semi-centralized	High
Processing Speed (TPS)	High	Fair	High	High	Medium	Low
Energy Cost	Very Low	High	Low	Low	Low	High
Security	Low	High	High	High	Low	High
Authenticity	High	High	High	High	High	High
Scalability	High	High	High	High	Medium	Low
Performance/ Efficiency	Very Fast	Fast	Very Fast	Fast	Fast	Slow
Consistency	Fork	Probability of forking	NA	NA	Fork	Fork
Adversary Tolerance	<51% validators	NA	NA	<25% computing power	<51% stake	<25% computing power
Application	BitShares, EOS	Dash, Decred, Es-pers, Peercoin	PoA, Ethereum, VeChain (VET)	Hyperledger Sawtooth	Voting, Casper, Cosmo, Ethereum, Nextcoin	Cryptocurrency, Bitcoin, Ethereum
Limitations	Moderate central-ization, prone to attacks	Hardware depen-dence, vulnerable to double spend-ing attack	NA	Equipment specific (Intel), lack of consensus finality, prone to forks	Validators with large stakes could lead to centralization, less secure than PoW, lack of consensus finality	Energy-intensive, high computational cost, vulnerable to forks and 51% attacks

DPoS = Delegated Proof of Stake, PoA = Proof of Activity, PoAu = Proof of Authority, PoET = Proof of Elapsed Time, PoS = Proof of Stake, PoW = Proof of Work

TABLE 5. Comparison of voting-based consensus algorithms.

Property/Feature	PBFT	Ripple
DLT Type	Permissioned	Permissioned
Decentralization	Low	Low
Processing Speed (TPS)	High	High
Energy Cost	Low/ Medium	Low
Security	High	High
Authenticity	Low	High
Scalability	High	High
Performance/ Efficiency	Fast	Very Fast
Consistency	No Fork	NA
Adversary Tolerance	Less than or equal to 33% faulty replicas	Less than 51% nodes in UNL
Application	Hyperledger, military, law enforcement	Ripple (XRP)
Limitations	Inefficient for large networks, prone to DoS and Sybil attacks	If the UNL is compromised, the system fails

PBFT = Practical Byzantine Fault Tolerance

TABLE 6. Classification of smart grids studies.

Research Focus	Study
Authentication and Autho-rization	[105], [106], [107], [108]
Access Control	[109], [110], [111], [112], [113], [114], [115]
Design Principles	[34], [116], [117]
Multi-agent Grids	[118], [101]
Energy Trading	[119], [102], [120], [121]
Consensus Mechanism	[122], [102]

In one of the review articles, Otoum and Mouftah discuss design principles that enable trustworthiness, secure energy trading, and remote monitoring in smart grids [123]. They perceive future smart grids at the intersection of artificial

intelligence (AI), federated learning, and blockchain. The blockchain-steered P2P energy network Brooklyn Microgrid (BMG) marked the world's first-ever energy transaction on the blockchain. It employs Ethereum-supported smart contracts and PBFT consensus algorithm to enable pro-sumers to sell their excess energy. Green Wallet (Germany) rents residential storage systems and uses blockchain to generate surplus energy from renewable sources. Verdigris Technologies provides a cloud-based solution to consumers that utilizes AI for optimal energy consumption. Verv is an AI-steered system in the UK that allows customers to track their appliances' energy consumption by itemizing energy costs. PowerScout is another solution. It partners with Google and the US Department of Energy to predict cost and energy consumption for smart homes by utilizing AI and machine learning.

Many researchers have used traditional cryptography schemes to manage trust in smart grids, such as Elliptical Curve Cryptography and Elliptical Curve Digital Signature Algorithm (ECDSA) or Secure Hash Algorithm 256 (SHA-256) [111], [119], [124]. Zhang et al. propose a keyless scheme to enforce trust between smart meters (SM) and service providers (SP) without requiring a trusted third party (TTP) [111]. SHA-256 algorithm and elliptic curve multiplication generate private and public keys for service providers. These keys are stored in the signature device instruction (SDI). The service providers and smart meters are registered and verified in the service provider network (SPN) through a hash function and public keys. Zhang et al. employ Shamir's Secret Sharing (SSS) algorithm for a safe power resource-sharing network [124]. A new Proof of Random Trust (PoRT) consensus mechanism is proposed to select a

leader node based on a dynamic random trust value. The leader selects validators according to the user information. Then, signs and broadcasts are sent to the power distribution node, the validator list, and the current blockchain height value. The height is encrypted by a power distribution node using the public key of each verification group member. The leader encapsulates the selected transactions into blocks and broadcasts them to the consortium chain. Muzumdar et al. used the Elliptic Curve-based Digital Signature Algorithm (ECDSA) to define a secure energy trading method [119]. A prosumer does energy trading with the smart grid via an energy injection contract and signs this transaction using ECDSA. On the other hand, an electricity consumer asks for energy consumption by calling the bidding contract. Mapping of supply and demand is performed via the trading contract in which all consumers' bids are collected and sorted. The bidding contract carries out the bidding recursively using the Vickrey auction method. Following the bid, cash coins are given to prosumers, and energy tokens are issued to consumers. The utilization contract verifies energy tokens' ownership with the customer's ID, and the energy is released to the consumer from the smart grid. After executing each contract, the blockchain is updated through Proof of Stake (PoS) consensus.

The idea of public and private key pairs along with digital signatures is demonstrated in [36], [105], [106], [107], [110], [112], [125], and [126]. Li et al. utilize the intrinsic property of blockchain to develop trust among electricity consumers [110]. The proposed architecture does not rely on a central miner; each user maintains its public and private key pairs. A user registers in the blockchain using distributed application software (DApp). A micro load management device (M-LMD) on the grid infrastructure layer periodically measures the load of the micro area. It sends it to a local grid aggregator (LGA) on the edge computing layer. A cloud server acquires all the data from all the concerned LGAs, and the dynamic electricity price is calculated and written into the blockchain. At the last blockchain overlay layer, privacy-sensitive data is encrypted by a public key and is stored, processed, and exchanged. The secure energy trading solution by Dorri et al. enables consumers and producers to directly send the message into the blockchain and negotiate the energy price by employing a destination public key routing [125]. Energy trading in the system is distributed by nature because of atomic metatransactions. Each smart meter has a public/private key pair, based on which it generates a Merkle tree; the signed root of the Merkle tree is used as a "certificate of existence." Lastly, the miners verify the transaction by matching the public key hash in the Merkle tree with the signature on the transaction. Patsonakis et al. defined a virtual node layer between customers and the aggregator to reinforce trust in smart grids [126]. The fog-enabled intelligent devices (FEIDs) are clustered in virtual networks using optimal profiling and segmentation. Here, the aggregator is enhanced with a sophisticated decision support system for self-balancing and grid stability. The privacy and

anonymity of transacting parties are ensured by employing standard encryption and built-in identity mixing schemes based on zero-knowledge proofs. Wang et al. introduce an authentication and key agreement protocol for smart grids based on edge computing [105]. The registration authority is responsible for registering the end user (a smart meter). As the initial step, a blockchain is established by creating a genesis file, including the configuration parameters. The smart contracts deployed here manage the key materials table, providing conditionally anonymous authentication without the asynchronous problem. Another research defines a lightweight authentication protocol for smart grids with the name BlockSLAP [106]. All the smart grid network nodes upload their hashed IDs to a nearby registering authority. The novelty is that the system parameters are incorporated in smart contracts and are visible only to the contract initiators. The smart contracts automatically generate public keys and digital signatures according to the pre-defined rules. At the final stage, the smart grid and the utility center authenticate each other and negotiate a session key for bidirectional secure communication. Zhong et al. addressed the issues of illegal access to smart grid entities by designing a blockchain-enabled authentication and authorization protocol [107]. Each user in the smart grid network is identified by a public key uploaded to the blockchain and publicly visible to everyone. A user signs into the system using a secret key. The service provider verifies the login request and urges the user to sign it. The user generates a token signed by their private key. Finally, the service provider verifies this token and decides on the user's login authorization. Timestamps are used to authenticate each transaction record, and it is ensured via signature that the contents of a transaction are not modified. The performance efficiency of the devised protocol is demonstrated using the Financial Services Blockchain Consortium (FISCO) consortium platform with an embedded PBFT consensus algorithm. In a similar attempt, Bera et al. utilized pairwise secret keys to design an access control mechanism to secure communication between smart meters and service providers [112]. For that matter, a smart meter mutually authenticates with its associated service provider through the "node authentication" process and establishes a session key among them through the "key establishment" process. The pairwise secret keys among the service providers are used for the secure consensus procedure. The service providers securely collect the data from their respective smart meters before forming the blocks. The blocks are added to the blockchain through a voting-based PBFT consensus mechanism in the P2P service provider's network. Hussain and Farooq introduce a cloud-based framework to build trust between the distributed system operator (DSO) and smart meters [36]. A smart meter uses public key cryptography to enter the DSO's blockchain network. The SM sends power computation data (PCD) to the DSO regularly, and the DSO aggregates PCD by using the private key of the DSO and validates it by applying the Clique PoAu consensus algorithm. The DSO approves the energy transactions,

which are transferred to the blockchain in the AWS cloud server.

Samuel et al. [118] and Khalid et al. [127] introduced the concept of direct trust (DT) and recommended trust (RT) to define decentralized trust management systems. The trust management system for multi-agent systems (MAS) by Samuel et al. follows a four-layer architecture [118]. The topmost blockchain layer stores all trust values calculated by the aggregators and encrypts feedback from multiple agents. The lower layer of the aggregator layer runs an incentive mechanism that enables a participating agent to perform DT and RT evaluations of other agents. Here, a weighted average trust method calculates the trust values. The upper layer performs the trust credibility of agents regarding trust consistency, distortion, and reliability. Agents produce, consume energy, and exchange trust evaluation values at the prosumer layer with aggregators. At the lowest physical layer, energy utilization information is exchanged between agents. Khalid et al. utilize Publicly Verifiable Secret Sharing (PVSS) to protect sensitive information from unauthorized participants [127]. A leader packs the selected transactions into blocks comprised of trust evaluation collection, trust uploading, miner selection, and block creation. In the initial phase, each agent's trust values are generated and checked for trust credibility using trust consistency, distortion, and reliability. The credibility of each agent is dependent on its cooperation, and a tit-3-for-tat (T3FT) repeated game strategy is developed to encourage cooperation between the agents.

Dehalwar et al. demonstrated the use of certificate generation for managing trust among IoT devices operating in a smart grid system [108]. In this study, a central authority is appointed to create a new block within the blockchain network for every Internet of Things (IoT) device (such as a smart meter) by authenticating its identity. The device transmits the relevant information to the block, and all the transactions are documented within the designated block. The grid operator (service provider) checks the validity of energy at each node and transaction by verifying the hash key (Merkel Root). Moreover, to retrieve the desired information from the new device, the service provider checks the device's authenticity from the registration authority.

Prabhakar and Anjali created two smart contracts in a consortium blockchain to control access and regulate energy trading among authorized parties [102]. Here, the AccessContract defines the access policies, and every access transaction is recorded in the blockchain for transparency by employing a modified version of the token-based scheme FairAccess. Whereas TradeContract is designed for efficient energy trading. After trading, an EnergyTraded transaction is recorded in the blockchain, and the net energy available for sale is updated. The authors also propose a new trust-dependent consensus protocol (TCO) that is resilient to one-third of the Byzantine voting power.

Jiang et al. used only a hierarchical consensus algorithm to strengthen trust management in smart grids [122]. The proposed framework includes an intra-layer algorithm,

an inter-layer algorithm, and a trust update mechanism. The smart grid nodes are divided into candidate, follower, leader, and key nodes. The voting mechanism is addressed using the tenure system and timer mode of the Raft algorithm and is implemented in the intra-layer. At the same time, PBFT is utilized for inter-layer consensus. The trust values of the nodes are updated based on the Eigen trust model.

Table 7 presents the comparative analysis of the past research work carried out in the energy sector to enforce trust with blockchain-based solutions.

VI. BLOCKCHAIN-BASED TRUST MANAGEMENT IN HEALTHCARE

This section addresses **RQ4** by surveying the literature on trust management models implemented through blockchain technology in the healthcare sector.

Blockchain-inspired healthcare systems provision for secure smart healthcare services. Besides this, substantial issues related to unauthorized access to electronic health records are introduced, and trust among the stakeholders is compromised [144]. Over several years, different paradigms have been defined to enhance the privacy and security of patients' data. A broad overview of the most influential contributions analyzed is reflected in Table 8.

In one of the surveys, Anjum Khurshid argues that blockchain's distributed structure and privacy-preserving features can be used to establish trusted systems for addressing public health needs [145]. One of the solutions is Rapid Supplier Connect by IBM, which is used to manage medical supply chains and detect counterfeits. The other IBM's Trust Your Supplier software provides stakeholders with a single point of transparency, visibility, and risk mitigation. BeepTrace, a mobile-based system developed at the University of Glasgow, provides encrypted and anonymized personal information to concerned authorities. Other applications of BCT may include patient identification or insurance approvals. Rahmani et al. see prospects of incorporating blockchain technology into cloud-based solutions to develop resilient trust models. The presented article highlights the trust challenges in centralized computing, such as huge overheads, trust evidence, less adaptiveness, and inaccuracy [143]. As cloud centers' data is stored at various locations at varying virtualization speeds, full physical or virtual server transparency is not provisioned. Furthermore, many decentralized models often require a third-party trust center or a certificate center. This results in several security challenges, such as high computational power and network overload.

The studies highlight the use of Elliptical Curve Cryptography (ECC) to manage trust in implementing smart healthcare [129], [142], [146]. The authors introduced a trust management framework for the Internet of Health Things (IoHT) [129]. Each IoT object generates a public-private key pair using elliptic curves (ECC). The Ethereum blockchain verifies the authenticity of IoT devices and groups their

TABLE 7. Summary of trust management in smart grids.

Study	Year	Trust Management Technique	DLT/Consensus Algorithm	Performance Metric (s)
Zhang et al. [111]	2019	Key generation using random numbers, SHA-256, elliptic curve multiplication	Consortium, Go Ethereum (Geth 1.7.2)	Storage cost, computational time, total time complexity
Li et al. [110]	2019	Public and private key pairs	Ethereum	-
Dorri et al. [125]	2019	Destination public key, atomic meta-transactions, private/public key pair	Public, Distributed Time-based Consensus (DTC)	Cost for creating accounts, success rate in tracking users
Patsonakis et al. [126]	2019	Identity mixing schemes based on zero-knowledge proofs, digital signatures	Hyperledger Fabric, BFT	-
Samuel et al. [118]	2020	Direct Trust (DT), Recommended Trust (RT)	Consortium	Trust values
Wang et al. [105]	2020	Master private key, master public key	Hyperledger Fabric, PBFT	communication cost, computation cost
Wang et al. [106]	2020	Pair of public key and digital signature, session key	-	Computation overhead
Zhang et al. [124]	2020	SSS encryption algorithm, random trust values	Consortium, PoRT	Liveness, security, scalability
Khalid et al. [101]	2021	DT, RT, T3FT for cooperation, PVSS for privacy	Consortium, PoC	Trust credibility, agents' cooperation and privacy
Muzumdar et al. [119]	2021	Elliptic curve-based digital signature algorithm (ECDSA)	Ethereum, PoS	Transaction throughput, energy injection time, energy trading time
Zhong et al. [107]	2021	Public key, secret/private key, signatures	FISCO, PBFT	TPS, registration time, login time
Bera et al. [112]	2021	Pairwise secret keys	Private, PBFT	Translation time, computation time
Dehalwar et al. [108]	2022	Certificate generation, device verification by miners (60%)	Hyperledger	Validation of nodes in log(n) time
Prabhakar and Anjali [102]	2022	Access control (AccessContract and TradeContract)	Hyperledger Fabric 1.1.0, TCON	Transaction validation probability
Jiang et al. [122]	2023	Trust-based hierarchical consensus	Consortium, Raft, BFT	Consensus efficiency, throughput, storage
Hussain and Farooq [36]	2023	Digital signatures	Ethereum, Clique PoAu	Block adding time

TABLE 8. Classification of smart healthcare studies.

Research Focus	Study
Security and Privacy	[128], [129], [130], [131], [132], [133], [134], [135], [136], [74]
Secure EHRs	[137], [138], [32], [139], [140], [141], [142], [55], [135]
Hierarchical Trust	[142]
MSN	[142]
Constrained Resources	[143]
Consensus Mechanism	[121]

identifiers into trusted zones. Association and aggregation are two types of interactions carried out by IoT devices in a trusted zone. Both communications are secured by Elliptic Curve Cryptography (ECC). Association requests are restricted to the same trusted zone, and the information between the zone members is exchanged based on smart contracts. On the other hand, the devices in one zone interact with devices in other zones using the aggregation request. Majdoubi et al. eliminate the need for a TTP by combining blockchain, Short Authenticated Strings (SAS), and ECC [146]. Here, the Ethereum blockchain stores the node identity with its associated public key and the digital signature, the SAS scheme (PV-SAS-MCA) authenticates network entities, and the lightweight ECC ensures efficiency. The whole authentication process requires less than 90 milliseconds. To address the trust issues in outsourcing EHRs to

a cloud server, Tan et al. introduce a two-sided verifiability scheme based on ECDSA [141]. The health records are encrypted using Searchable Symmetric Encryption (SSE) and stored in IPFS. Meanwhile, digital signatures are used to verify the authorized medical entities. In the data pre-processing stage, a bitmap index ensures searchability over encrypted Electronic Health Records (EHR). This index is created by the third party, which is replaced in this scenario with blockchain technology to verify the authenticity of the ECDSA public key. Additionally, all the health records are indexed and searched using blockchain. Bhan et al. define a clustered hierarchical trust management system (CHTMS) that is resilient against Denial-of-Service (DoS) attacks [142]. The methodology works in three phases: computation of trust at the node, computation of trust at CH, and calculation of trust at the base station (BS). The nodes in the network compute their digital signatures using the ECDSA, and public and private keys for each node are generated with ECC. Each node calculates the trust values for every other node and assigns one of the three possible states (trusted, untrusted, and uncertain) based on their trust value. Afterward, temporal data from each node is gathered, and the cryptographic hash value is computed using the secure hash algorithm (SHA). Following that, the trust state of all the group nodes is transmitted to a CH installed at the individual departmental level. If the CH recognizes any malicious nodes, it alerts the BS located at the hospital

level. The BS, in turn, verifies or refutes the malicious node.

Tahir et al. present a hybrid authentication and authorization framework using lightweight encryption for healthcare informatics [131]. A certificate authority creates certificates and keys for each healthcare user. The user submits a certificate, key, and data to the blockchain network. The chain verifies the user's certificate and key, formats the data, and sends it to a peer-to-peer cloud storage platform, which generates a hash code of the data and sends it to the blockchain for mining using Proof of Auction (PoAu). Miners are rewarded for successful mining, and their rating is degraded otherwise. Along similar lines, Malamas et al. demonstrated an authorization framework based on blockchain to manage fine-grained access to IoMT devices and data [147]. The peer-to-peer cloud storage platform stores patient health records in a distributed manner. Three smart contracts are implemented to manage varied access to medical devices: registration, actor handling, and log management. A new consensus algorithm, Proof of Medical Stake (PoMS), is designed to verify transaction logs with private and public keys. Furthermore, Mondal et al. present a trusted EHR management system using blockchain multi-signature stamp [140]. The framework also involves the concept of non-repudiation. A user cannot deny communication as hash links save the signature with a timestamp and add it to the immutable blockchain. User identities are created by the certificate authority and are validated by the membership service provider. Private keys are used to sign a block, and public keys are used to verify that signature. Selvarajan and Mouratidis formulated another trust management approach in which cryptographic keys are generated using a consultative transaction key mechanism. It combines blockchain, consultative key generation, Burrows–Abadi–Needham (BAN) logic, feedback analysis, and trust score calculation [74]. A patient requests a key from the database for authentication, and the system validates the trust of the requested patient through the Tuna Swarm Optimization (TSO) algorithm that performs nonce message verification. The trust is calculated using the feedback data: number of rejected requests, IP addresses, and websites. These rejections determine the trust score. A low trust value is indicative of a forbidden transaction. The BAN logic enables the smooth flow of information and authenticates the user for further access.

The research works reflect the usage of fuzzy logic in trust management for IoMT [134], [148]. Zulkifl et al. proposed FBASHI using three fuzzy inference systems [134]. A requesting entity (for instance, a doctor) initiates a transaction with specific parameters. A set of endorsing peers with assigned weights are nominated through PBFT. The endorsing peers verify these transaction parameters, and in return, they run the chain codes and verify the inputs and outputs. Gossip protocol is run so that all the peers of the concerned channels update their ledgers. Upon successful

transaction completion, an event is created, and the requesting node is granted access. Consequently, the service provider generates a Nonce to the requesting entity, also recorded on the blockchain. Another trust management scheme based on fuzzy logic was devised by Ali et al. [148]. The trust management module is implemented at the fog layer of the IoMT network. At first, a medical device enters the IoMT network, forming device clusters. Cluster heads are elected from the multiple devices in the network. In the later step, trust parameters are collected from each node, and the trust value against them is computed. All the trust values are stored in the blockchain. If the calculated trust value is less than the pre-defined threshold, it is declared a Sybil node (malicious) and isolated from the network. Lastly, the decision of a malicious or benign node is propagated throughout the network.

Bayesian inference is one of the statistical techniques that is used to define trust management systems. Khalfaoui et al. combine probabilistic Bayesian classification and stochastic Markov modeling to overcome insider and on-off attacks in healthcare services [130]. Firstly, the network nodes' behavior is studied using a probabilistic model based on a time-homogeneous Markov chain. Then, a stochastic approach is used to classify them into honest, suspicious, and malicious based on their trustworthiness. The probabilities of trustworthiness are computed using a Bayesian classifier. Permissions are successively updated using attribute-based access control (ABAC). If the prediction shows the node as honest, it continues benefiting from its current privileges, and if it is detected as malicious, it is banned from the network. Meng et al. also exploited Bayesian inference to deal with insider attacks in the Medical Smartphone Network (MSN) [149]. The blockchain trust management system is divided into two layers to overcome the single point of failure. The MSN layer handles the interaction between MSN nodes and the central server. Meanwhile, the Chain layer constructs a consortium blockchain that allows each node to upload features of malicious packets. As each node can access the chain to examine the features of malicious packets, they quickly update their blacklist and immediately halts the interaction with the malicious nodes. The research work by Li et al. regulates trust management in an MSN [150]. The central server is responsible for managing the network, and this includes measuring the reputation of each MSN node based on Bayesian inference, performing adaptive traffic sampling, and retrieving the verified data from the blockchain. The status of incoming traffic in the network is monitored by installing an Intrusion Detection System (IDS) in the collaboration component of an MSN node. Moreover, the chain component is designated to append the information of malicious traffic to the chain. The updated chain is shared within the blockchain network, and the other nodes verify the malicious payload.

Jeon et al. introduce a novel pattern-based scheme to curb data falsification, inconstancy, and misuse in collaborative

healthcare setups [135]. The trust model prepares the basis of information protection through data encapsulation and encryption. Six attributes are organized to build trust and are applied as the trust model verification criteria. The elements of trust in the collaborative business processes are sorted into five risk areas. The trust elements, i.e., trust concern and uncertainty, are deduced from the risks in a business process and subsequently sorted as a pattern using BCT. Each trust pattern is mapped with trust elements.

Table 9 gives an overview of the trust management techniques devised for smart healthcare.

VII. BLOCKCHAIN-BASED TRUST MANAGEMENT IN TRANSPORTATION

This section addresses **RQ4** by surveying the literature on trust management models implemented through blockchain technology in VANETs. In the past few years, blockchain technology has positively impacted the transportation sector in numerous ways, hence the name smart transportation. Blockchain-based systems are being researched with a keen focus on autonomous and connected vehicles [167]. The reliability and security of the vehicle-and-roadside collaboration directly affect transportation safety, as many onboard devices and roadside infrastructures communicate for information exchange [168]. Trust management systems play a crucial role in protecting the privacy, preservation, and identity protection of vehicles and their data.

Table 10 depicts the classification of research papers centered on trust management in smart transportation.

Wang et al. design a privacy-preserving trust management system that leverages ECC to evaluate the trustworthiness of vehicles [169]. Two smart contracts are designed to achieve the purpose, i.e., feedback aggregation smart contract (FASC) and trust evaluation smart contract (TESC). The roadside units collect all the transactions among the active vehicles and generate a new block by combining PoW and PoS consensus protocols.

Another cryptography technique is employed by Kudva et al. [170]. During registration, a roadside unit assigns a vehicle with a public and private key pair using the Elliptic Curve Diffie-Hellman (ECDH) algorithm. When joining the network, neighboring vehicles calculate their trust values individually using the modified Ad hoc on-demand distance vector (AODV) routing protocol. In the second phase, authorized roadside units (RSUs) produce aggregated trust scores for all the vehicular nodes. Consequently, the denylist node tables are dynamically modified, and the leader RSU is selected via PBFT consensus to generate a block in the blockchain.

Similarly, Malik et al. introduced a two-phase trust mechanism for a Vehicular Ad hoc Network (VANET) [171]. To send a message within the network, a vehicle must first request a unique encryption key from a nearby roadside unit through a formal request process. All the keys are maintained on the blockchain ledger and are optimally tuned with the Sea Lion Explored-Whale Optimization Algorithm (SLEWOA).

At the sink, the receiver requests the corresponding RSU for the key, and the RSU makes node trust ability prediction using a rule-based and neural network-based evaluation.

Later, Liu et al. combined trust management with conditional privacy-preserving announcements for VANETs [172]. At first, a vehicle registers itself with the trusted authority (TA) anonymously and sends messages in a non-trusted environment to perform threshold authentication. If a malicious vehicle sends a message, the TA traces its identity and matches it with the identity-based key and the public address (pseudonym) stored in the blockchain. The RSU calculates the messages' reliability using logistic regression based on vehicles' reputation values. The resultant reputation value is used to determine whether reported messages are correct. The RSU packs the updated reputation data and adds this block to the blockchain using a mix of PoW and BFT consensus algorithms. Lastly, the announcement protocol achieves conditional privacy by exploiting the identity-based group signature scheme.

Based on Bayesian inference, Yang et al. proposed using blockchain as a means of decentralized trust management for vehicular communication [8]. First, all vehicles assess the credibility of received messages and then generate ratings. The positive rating indicates a credible message, while the negative rating represents a false one. These ratings are into one data block and uploaded to a local RSU. Based on vehicle ratings, the RSU calculates the offset of trust values for every vehicle. The trust value offset is positively correlated with the ratio of positive ratings. The sum of all offsets is the trust value of this vehicle. A given RSU competes against other RSUs in the network via a joint PoW and PoS consensus algorithm to determine which will be elected as the miner. This prevents RSUs with high ratings (stakes) from always uploading to the chain and allows less common RSUs to upload their data to the blockchain.

Zhang et al. formalize a trust management mechanism that uses blockchain to save reputation values of vehicular nodes [173]. A reputation value update algorithm includes message credibility calculation, rating mechanism, and reputation value calculation based on ratings. This algorithm detects vehicles that send false messages; hence, their reputation value is reduced. A consensus mechanism that combines PoW and PoS is designed to update the vehicles with changes in reputation value. To authenticate each message, the information credibility and the information credibility calculation methods are defined. The vehicle summarizes all messages based on Bayesian inference. Two types of rating mechanisms, the initial rating and the final rating, detect and penalize vehicles that send malicious messages. After collecting all the ratings, an RSU uses a weighted aggregation algorithm to calculate the reputation value of the rated vehicles based on their reputation values.

Various studies propose the concept of multi-dimensional or multi-layered trust in connected vehicles [6], [165], [174], and [175]. Kandah et al. address the challenges of trust management for connected vehicles by proposing a multi-tier

TABLE 9. Summary of trust management in smart healthcare.

Study	Year	Trust Management Technique	DLT/ Consensus Algorithm	Performance Metric (s)
Malamas et al. [147]	2019	Private and public keys	Ethereum, PoMS	Integrity and traceability
Khalifaoui et al. [130]	2020	Markov chain, Bayesian classifier	Ethereum testnet	Energy consumption, execution time
Abou-Nassar et al. [129]	2020	ECC	Ripple	-
Tahir et al. [131]	2020	Lightweight encryption, random numbers, joint conditional probability	Ethereum, PoAu	Authenticity, access control, communication cost, computational overhead
Meng et al. [149]	2020	Bayesian inference	Consortium blockchain	Average CPU workload
Li et al. [150]	2021	Traffic sampling-based, Bayesian inference	-	Trust values
Mondal et al. [140]	2021	Multi-signature stamp	Hyperledger, common channel consensus	Confidentiality, integrity, authentication
Majdoubi et al. [146]	2022	SAS, ECC	Ethereum	Execution time, authentication time, CPU power consumption
Zulkifl et al. [134]	2022	Three fuzzy inference systems	Hyperledger, PBFT	Transaction latency, throughput
Tan et al. [141]	2022	Searchable Symmetric Encryption (SSE), Cipher-Block Chaining (CBC) encryption mode of the Advanced Encryption Standard (AES), Elliptic Curve Digital Signature Algorithm (ECDSA)	Ethereum	Indexing time, encryption rate
Selvarajan and Mouratidis [74]	2023	Consultative Transaction Key Generation and Management (CTKGM), Tuna Swarm Optimization (TSO)	Hyperledger	Key generation time, encryption time, decryption time, entropy level, computational efficiency
Jeon et al. [135]	2023	Pattern based on trust elements	Ethereum 2.0, PoAu	Integrity, security, risk elements
Ali et al. [148]	2023	Fuzzy logic	-	Energy consumption, attack detection, trust reliability, packet delivery ratio, throughput
Bhan et al. [142]	2023	ECC	Hyperledger Fabric v2.2	CPU utilization

TABLE 10. Classification of smart transportation studies.

Research Focus	Study
Privacy Preservation	[151], [152], [153], [154], [155], [156], [157]
Secure Communication	[158], [159], [160], [161], [162], [163], [164]
Multi-dimensional Trust	[156], [153], [45]
AI-enabled Trust	[165]
Constrained Resources	[166]

blockchain-based method [6]. Each vehicle in the system maintains an information table that is continuously updated, and three types of trust are incorporated into the proposed system to ensure consistency among vehicles and deliver the most accurate trust factor for any given vehicle in the system.

Zhang et al. propose an AI-enabled decentralized trust management model [165]. As the first step, each vehicle generates and exchanges messages with other vehicles. Once a vehicle receives messages from a nearby vehicle, it computes the sender's local trust level (LTL) using a feedforward neural network. Afterward, the global level trust (GTL) of all the participating vehicles is calculated by the RSU. The GTL is encoded and added as a new block to the blockchain. During the last stage, global trust-level voting and dissemination by all the RSUs are carried out.

Another trust management scheme suggested by the authors relies on three types of trust: detection trust, reference trust, and transmission trust [174]. A vehicle continuously sends detection probes to all the surrounding

nodes, determines if its neighbors have abnormal behaviors, and calculates the direct trust between them. When the entire detection process is done, the vehicle uploads the associated trust values onto the primary servers. After the primary servers update the BlackList field of the blockchain and push it to RSUs, RSUs then forward the status to their surrounding nodes. When the servers receive the new trust values of nodes, they update them based on the history values stored in the blockchain and generate a new block. Wang et al. define another multi-layered trust management scheme for intelligent vehicular collaboration in the next-generation networks (6G) [175]. Combining two consensus algorithms, PBFT and PoS, improves scalability and reduces computational complexity. The blockchain deployed on the edge aggregates all the reputation data and stores each node's updated global trust values in the block. Global trust is calculated by taking the average of all the evaluating nodes' local trust values; most trustworthy nodes are assigned more significant weight. A node with a high reputation (global trust) is considered more trustworthy.

Other miscellaneous techniques were proposed over time. The blockchain-based trust management framework devised by Theodouli et al. relies on World Wide Web Consortium (W3C) standards for Decentralized Identifiers (DIDs) and Verifiable Credentials (VCs) [176]. The identities of the vehicles are stored in a decentralized fashion, which avoids a single point of failure and eliminates the need for a centralized Certificate Authority (CA) to manage the identifiers.

Cinque et al. formulated a trust management system for IoV using Libelium Wasp mote and the ZigBee protocol [177]. The blockchain is implemented at the fog layer to record transactions among the vehicles. Each block contains the trust values associated with ledger entities. The ledger entities host the blockchain and respond to the Trusted Node (TN) requests. The nodes periodically inform the IoT devices about the host trust degrees.

Cong Pu suggests another trusted communication mechanism based on blockchain [178]. Whenever a vehicle detects a traffic-related event, it immediately generates and broadcasts a message to the nearby vehicles. Additionally, it evaluates the credibility of the received message, generates the trust value of the message originator, and uploads the trust value to a nearby RSU. The RSU calculates the reputation value of the message originator by using a multi-criteria decision-making model, packs the reputation value into a block, and adds the block into the blockchain. Once a block is added, other vehicles can access each vehicle's reputation value whenever needed.

Dewanta and Mambo facilitate fog computing service among vehicles without network infrastructure [179]. The local trust is based on the bidding price transaction scheme between client and server vehicles. At the same time, the global trust is a payoff assignment based on transaction evaluation. Certified public credentials (CPC) and certified secret credentials (CSC) are assigned to all vehicles during registration with the trusted agent on the fog layer. While passing through a rural area, vehicles establish local trust among them, performing authentication, session key generation, and certified transaction record (CTR) checking. At the end of the session, both CRT and offloaded data are stored in the fog node for verification and authentication to provision secure communication among the vehicular nodes.

Liu et al. use the Hidden Markov Model (HMM) to define blockchain-assisted trust management [180]. Three layers, viz; network layer, edge-blockchain layer, and certificate authority (CA) layer, constitute the system design. Here, unsupervised learning is used to train the model because the behavior of the vehicles is never anticipated.

Inedjaren et al. propose a fuzzy logic approach to improve trust among vehicles [181]. The vehicles are inducted into and isolated from the blockchain network based on the vehicles' trust value. A malicious vehicle is detected by fault tolerance optimized link state routing (FT-OLSR); a node with a trust value less than or equal to a certain threshold is considered malicious. On the other hand, nodes with trust values more significant than the threshold are qualified as reliable.

A new Proof of Trust (POT) consensus algorithm that is highly effective in resource-constrained VANET environments is introduced by Singh et al. [182]. A vehicle generates its temporary public and private keys for registration. The trusted authority issues the certificate to a vehicle to enter the network after the endorsement from the certificate authority (CA). The RSUs receive all the transactions corresponding to the global blockchain and aggregate the received transactions

together to form a block and deploy the block on the blockchain using the PoW. When a vehicle wants to exit a particular region, the regional authority executes a global transaction against the permanent public address of the vehicle. It updates the score values on the global blockchain by taking the current trust value and wallet score.

A hardware primitive-based assignment of unique identifiers to vehicles is introduced in the research by Javaid et al. [183]. The trust mechanism is based on the blockchain public-key infrastructure (PKI). The Physical Unclonable Function (PUF) assigns a unique identifier to each vehicle in the IoV network through which trust is established. Each vehicle holds its blockchain account and public and private keys for secure encrypted communication. Here, the Delayed Proof of Work (dPoW) appends a block to the blockchain by differentiating between registered and malicious vehicles.

A summary of the literature review for trust management implementation in smart transportation/ VANETs is presented in Table 11. Table 12 summarizes the literature review conducted in three smart city applications: smart grids, healthcare, and transportation.

VIII. RESEARCH CHALLENGES AND OPEN ISSUES

In response to **RQ5**, this section outlines some key challenges and open issues that affect the operational implementation of blockchain-inspired trust management systems. These gaps in knowledge are identified to provide a clearer understanding of the research landscape.

Introducing blockchain for smart city operations presents pressing challenges related to resource collaboration and sharing, among security, privacy, usability, and wasted resources [167]. Despite increased interest in blockchain technology in academic and industrial circles in recent years, the security and privacy of this technology remain a topic of ongoing debate [25]. It can be adapted for applications like IoT, cybersecurity, robotics, and many other areas. However, some challenges are still associated with blockchain intelligent adaptability and efficiency [164]. The critical pitfalls of blockchain-enabled trust mechanisms are efficiency, robustness, scalability, and QoS, including sparsity, consistency, availability, and privacy concerns as some of the biggest challenges to be addressed [182]. The most common trust management issues in IoV are inefficiency, low latency, heterogeneous trust evaluation, and aggregations to the high dynamic nodes [175], [194].

Apart from this, some issues still exist when implementing blockchain for real-world security applications. For instance, it is sometimes necessary to link transactions and contracts to known identities in most blockchain-inspired applications [33]. This raises the privacy and security concerns of the data stored on the shared ledger. Additionally, BCT can become a target for cyber-criminals and subsequently suffer from various attacks like distributed denial-of-service attacks (DDoS). Others may include inaccurate profile establishment, limited signature coverage, and massive false alerts.

TABLE 11. Summary of trust management in smart transportation.

Study	Year	Trust Management Technique	DLT/ Consensus Algorithm	Performance Metric (s)
Yang et al. [8]	2018	Bayesian inference	PoW and PoS combined	Trust value offsets, unfair rating ratio, block generation time
Kandah et al. [6]	2019	Direct trust, indirect trust, reputational trust	Proof of Interaction (PoI), PBFT, Lightweight Mining Protocol (LWM)	Average number of blocks, time between miner election
Theodouli et al. [176]	2020	Decentralized Identifiers (DIDs) and Verifiable Credentials (VCs)	Consortium	Scalability and Velocity.
Malik et al. [171]	2020	Privacy preservation model, Node trustability prediction, Sea Lion Explored-Whale Optimization Algorithm (SLEWOA), rule-based evaluation, neural network-based evaluation	-	Rejection ratio
Liu et al. [172]	2020	Group signatures, logistic regression	PoW and PBFT combined	Average computation time, reputation value, false recognition rate
Cinque et al. [177]	2020	ZigBee protocol	Permissioned blockchain	Average query response time, battery discharge, robustness
Javaid et al. [183]	2020	Physical unclonable function (PUF)	Delayed Proof of Work (dPoW)	Computational overhead
Wang et al. [169]	2021	ECC	Hyperledger Fabric, PoS and PoW combined	Average trust value
Cong Pu [178]	2021	Multi-criteria decision-making	-	Fictitious message detection rate, malicious vehicle detection latency, number of dropped fictitious messages, average of trust values
Dewanta and Mambo [179]	2021	Certified Transaction Record (CTR)	-	Connection timespan
Liu et al. [180]	2021	Hidden Markov Model (HMM)	Hyperledger Fabric, CouchDB	Trust evaluation accuracy
Zhang et al. [165]	2021	Local trust using feed-forward neural network, Global trust	Merkle tree	Precision, recall
Kudva et al. [170]	2021	Elliptic-curve Diffie-Hellman (ECDH)	Consortium blockchain, PBFT	Packet Delivery Ratio (PDR), throughput rate
Inedjaren et al. [181]	2021	Threshold based	PoT	Detection rate of malicious vehicles, detection overheads
Singh et al. [182]	2021	Trust values, wallet score	Private-permissioned Ethereum, PoW	-
Li et al. [174]	2021	Detection trust, reference trust, transmission trust	-	Detection accuracy, delivery ratio
Zhang et al. [173]	2021	Reputation based, Bayesian inference	Ethereum, PoS and PoW combined	Reputation value, block generation time
Wang et al. [175]	2023	Multidimensional trust (direct and indirect)	PoS and PBFT combined	Message credibility, throughput, time delay

TABLE 12. Summary of trust management systems for smart cities.

TM Technique	Smart Grids	Healthcare	VANETs
Bayesian Inference	-	[130], [149], [150]	[8], [173]
Digital Signatures	[126], [36]	[140]	[172]
DT and RT, LT and GT, Trust Values	[118], [101]	-	[6], [165], [182], [174], [175]
Elliptic Curve Cryptography (ECC)	[111]	[129], [146], [142]	[169], [170]
Elliptic Curve Digital Signature Algorithm (ECDSA)	[119]	[141]	-
Fuzzy Logic	-	[134], [148]	-
Pair-wise Keys	[110], [125], [105], [106], [107], [112]	[147]	-
SHA-256	[111]	-	-
Shamir's Secret Sharing (SSS) Encryption	[124]	-	-
Miscellaneous	[108], [102], [122]	[131], [141], [74], [135]	[176], [171], [177], [183], [178], [179], [180], [181]

The survey analysis reveals that existing trust management models for smart city applications are limited in sophistication and face significant challenges in theoretical and practical contexts. The most common trust management issues in IoV are inefficiency, low latency, heterogeneous trust evaluation, and aggregations to the high dynamic nodes [175]. The following are some of the technical

challenges associated with using blockchain-based trust management mechanisms in a smart city environment:

- 1) *Privacy*: Blockchain provides exceptional resilience with cryptographically empowered security [195] as there exists no single point of compromise due to the technology's distributed nature. However, it does not include confidentiality among its primary objectives.

TABLE 13. Security Issues in Smart Grids, Smart Healthcare, and Smart Transportation.

Study	Block Veri- fication	Key Man- agement	Messages' In- tegrity/Secure Communi- cation	Mutual Au- thentication	Nodes Identifica- tion/Access Control	Smart contract vulnerabil- ity	Malicious attacks	Privacy Preser- vation
Smart Grids								
[111]	✓	✓	✓					
[105]		✓		✓				
[112]		✓			✓			
[107]			✓		✓	✓	✓	
[106]				✓			✓	
[101] [108]					✓			
[119] [122]								✓
Smart Healthcare								
[184]	✓							
[134]					✓		✓	✓
[141]					✓			✓
[185]							✓	✓
[132]								✓
[135]			✓		✓			✓
[140]								✓
[128]					✓			✓
[142]					✓		✓	✓
[74]		✓	✓					
[129] [139]			✓		✓			
[137]					✓			
Smart Transportation								
[45] [186]		✓		✓				
[127] [174]							✓	✓
[151] [154]								✓
[170]							✓	
[172]			✓					✓
[180]							✓	
[181] [187]			✓					
[182]				✓				
[188]		✓	✓					
[189]								✓
[190]				✓			✓	
[191]				✓				
[192]							✓	
[193]				✓				

One of the potential risks involves losing the private key, which may provide attackers with copies of databases, and the possibility that network validation processes may allow fraud through double expenditure or a transaction reversal.

- 2) *Security*: Cybersecurity challenges in BCT may hinder blockchain implementation in security and privacy applications [196]. Trust models are not robust enough to deal with certain advanced malicious attacks, especially malicious recommendations. Given the newness of blockchain technology, some vulnerabilities expose users to cybercrime. Wallet security attacks and 51% attacks are the most recognized blockchain security threats [27]. While others involve server breaches, illegal transactions, and wiretapping, to name a few. A 51% attack, also called a majority attack, is a potential threat to the integrity of a blockchain system in which a group of malicious miners takes over 50% of a network's mining hash rate—subsequently causing network disruption. Other security threats include long-range attacks, distributed

denial-of-service (DDoS), P + epsilon attacks, Sybil Attacks, balance attacks, and border gateway protocol hijacking (BGP) [99]. Table 13 highlights each selected paper's security issues.

- 3) *Consensus Protocol*: A consensus mechanism is a foundational aspect of BCT. While designing a blockchain-enabled trust management model, selecting the most suitable consensus protocol is a crucial task as it significantly affects the operation of the trust mechanism [180]. A consensus algorithm decides which node has the right to become a miner to validate transactions and generate a new block in the blockchain. Each consensus algorithm has varied capabilities to improve the blockchain's computational efficiency while ensuring data consistency and security [197]. Nevertheless, the traditional consensus algorithms have limitations in terms of requiring high computing power, slow consensus formation, and low data throughput. The limitations of each are highlighted in Table 4 and Table 5. Many customized algorithms specific to their application have been proposed by

combining the strengths of widely known algorithms or by incorporating divergent technologies, such as virtual mining technology. Different algorithms demonstrate distinctive characteristics, strengths, shortcomings, and trade-offs in different scenarios. Conclusively, the choice of a consensus algorithm depends on the specific application requirements to provide security, scalability, and energy efficiency.

- 4) *Adaptability and Interoperability*: Shifting from a conventional centralized trust management system to a blockchain-based decentralized system requires much technical effort. For instance, trust management seems less deployable with public blockchains primarily due to data heterogeneity issues. On the other hand, a permissioned blockchain has separate modalities for usage and implementation. One pressing challenge is integrating a trust management system within existing systems due to certain constraints. For instance, incorporating a trust mechanism into edge computing brings additional challenges mainly related to security as its deployment and computing capabilities occur distributedly. An edge node has limited computing and storage resources; this hinders efficient task execution and block generation. Conclusively, it compromises the resource allocation in edge environments. Furthermore, most trust value calculations do not adequately consider the trust value between smart devices and edge service providers when implementing trust management with edge computing systems [14].
- 5) *Trusted Third Party (TTP)*: Most trust management models are centralized, and even those that claim to be decentralized models still need a trusted third party or a certification authority most of the time. This poses miscellaneous security risks such as single point of failure, overloading, credibility loss, etc.
- 6) *System Overheads*: Huge overhead costs are associated with implementing an efficient trust management system. This includes but is not limited to resource utilization and computational complexity. This restricts the usage of sophisticated trust solutions in large-scale networks.
- 7) *Latency and Throughput*: Blockchain-enabled trust models are based on traditional consensus protocols, mostly like PoW. These are computationally intensive and slow, leading to high latency, which is unsuitable for dynamic networks like VANETs [193]. Alternative Consensus Mechanisms like PoS or PBFT can reduce latency but still may not meet the stringent low-latency requirements for real-time VANET applications. This also poses propagation delays, especially in large-scale VANETs with high mobility [181]. Additionally, conventional blockchains like Ethereum and Hyperledger have limited Transactions per Second (TPS) capability, which can be a bottleneck in high-demand VANET scenarios. As the number of vehicles increases, the network handles become impotent to manage

a correspondingly higher number of transactions, stressing the throughput capabilities [198].

- 8) *Data Immutability and Legal Ownership*: Blockchain's immutability property maintains the integrity and trustworthiness of the data [182], [199]. However, any fraudulent or erroneous data once entered into the blockchain, adversely affects the VANET operations and leads to error propagation throughout the network. Moreover, if the data needs to be corrected or updated, this immutability can lead to complexity [200].
- 9) *Governance Issues*: Establishing trust in a decentralized environment is itself challenging due to the involvement of numerous stakeholders [201]. In the case of VANETs, achieving consensus among vehicle owners and government agencies for regulatory oversight and coordination is hard.

IX. CONCLUSION

This survey overviews recent advances in blockchain-based trust management systems for smart cities. Blockchain technology can potentially revolutionize various sectors within smart cities, including transportation, healthcare, and energy management. However, the realization of these benefits relies heavily on the ability to establish trust among the various stakeholders involved. Traditional security architectures may struggle to address the resource constraints inherent in smart city environments, making blockchain-based trust management systems an attractive alternative.

The article analyzes 51 papers published from 2018 to 2023 that cover different aspects of blockchain, such as its definition, characteristics, classification, applications, challenges, and future directions. The survey has highlighted several key benefits of using blockchain technology for trust management within smart cities.

- 1) Decentralization is achieved by eliminating a single point of failure, which enhances security and resilience.
- 2) The immutability of blockchain transactions ensures that data cannot be altered or tampered with, providing an additional layer of trust.
- 3) The traceability of blockchain transactions prevents malicious use of data.

Despite these advantages, significant challenges and limitations are still associated with implementing blockchain-based trust management systems within smart cities. These include scalability issues, performance bottlenecks, and the need for further research into the practical applications of this technology. Nevertheless, the benefits of blockchain technology in enhancing trust management in smart cities are substantial, and further investigation is warranted to explore these opportunities fully.

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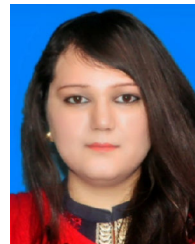
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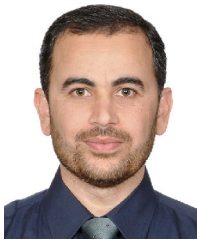
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